# Imran Khan

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## Education

2021–2025 : **Ph.D. in Computer Engineering**, *Northeastern University*, Boston, USA.

2018–2020 : **Masters of Science in Computer Engineering**, *Southern Illinois University Carbondale*, IL.

## Skills

Languages: Python, C++, Java/Android, Bash, SQL, HTML, CSS

Tools: XCAL-Accuver (5G/LTE signaling message analysing tool) || Keysight RuSIM || Spectrum Analyzer || Git || Wireshark || Scikit-Learn || PyTorch || Keras || SoA Machine-Learning Frameworks || Video Streaming tools (Puffer, FFmpeg)

Platform: Linux || Android || Docker || Cloud (AWS, GCP)

Protocols /Standards: NR/LTE 3GPP standards || TCP/MPTCP/UDP protocols and their implementation (Linux Source Codes) || IEEE 802.11 ax/ac/ad/b/g/n standards

## Experience

### Research Assistant at Institute for the Wireless Internet of Things

2021 – 2025 ***Conducted research on 5G and next generation of Cellular Networks***.

### Research Intern and External Collaborator at AT&T Labs. Inc

2021 – 2022 ***Investigated the impact on QoE of low-latency video streaming application over 5G networks***.

## Projects

***X5G: O-RAN compliant 5G Testbed Development***, *NSF Funded*.

OAI
C++
O-RAN
NVIDIA ARC

- Contributed to the development of X5G: an O-RAN compliant multi-vendor (NVIDIA, OAI, Foxconn) programmable testbed for research.
- Instrumented OAI code for performance tuning, configuration tweaking, and KPI extraction.
- Developed measurement report and event (A3) handling mechanism for Xn-handover.
- Improved system throughput by 20% by debugging and bug reporting to Nvidia Aerial and OAI team.
- Published 1 Conference and 1 Journal Paper.

***5G measurement studies on user-perceived performance***, *NSF Funded*.

5G/LTE
Python
X-CAL
Puffer
FFMPEG

- Designed and executed large-scale mobility measurement campaigns (multi-city/cross-country drives), integrating synchronized application-level tests with cellular KPIs for holistic analysis.
- Assessed 5G coverage, throughput, latency, and user Quality of Experience (QoE) using large-scale dataset [10TB+].
- Built an end-to-end measurement and QoE extraction pipeline for modern interactive applications, including AR/VR, 360° video streaming, and cloud gaming.
- Developed tools to extract lower-layer (PHY/MAC/RLC) KPIs from 5G control-plane signaling (RRC) [XCAL logs].
- Investigated key 5G mechanisms—handovers, uplink power control, and advanced features (carrier aggregation, beamforming, MIMO)—to identify performance bottlenecks and explain observed behavior.
- Built automated Python pipelines for data cleaning, feature extraction, and statistical analysis; produced publication-quality figures and reproducible artifacts.
- Open-sourced **datasets/scripts** and created replication-friendly workflows to enable follow-on studies.
- Published 8 conference papers.

***Throughput prediction in operational 5G networks***, *CISCO Funded*.

5G/LTE
Python
Machine
Learning

- Collected a fine-grained, real-world **dataset** during a 4-day (1000+ km) drive, measuring bidirectional application throughput at 100 ms time-scales across all three major U.S. operators and mixed LTE/5G deployments.
- Evaluated multiple predictors (MLP, GDBT, transformer-based MVTS) alongside statistical baselines (EWMA, last-value) and showed that low RMSE can still yield poor application-centric accuracy (e.g., high ARE95 / low PARE10).
- Used over-parameterized neural networks as a "learnability test" and found slow training convergence and persistently high validation/test error, indicating fundamental limits to fine-grained throughput predictability.
- Identified the root cause as conflicting sample sequences—highly similar recent histories followed by drastically different next-sample throughput—explaining why models must learn spiky functions and generalize poorly.
- Published 1 Conference paper.

***5G Performance Evolution Study***, *NSF Funded*.

JAVA
Android
Python
AWS
Machine Learning

- Built an IRB-approved **Android measurement app** (NextG-UP) to collect uplink TCP throughput and RTT alongside cellular KPIs (GPS, RSRP/RSRQ/RSSI, Cell-ID, ARFCN/EARFCN, operator); implemented nuttcp (10s UL test) + ICMP ping (11 packets @200ms) with a lightweight footprint (6.5MB APK, <250MB RAM).
- Deployed a geographically distributed measurement backend (AWS servers in N. Virginia, Oregon, Frankfurt + AWS Wavelength inside Verizon's network in Boston) and integrated nearest-server selection + mobility modes (static/walking/driving).
- Led a cross-sectional, year-long 5G uplink campaign with weekly volunteer measurements: 10,166 tests across 8 cities in 7 countries (2 continents), covering 12 operators and 1,422 unique cell IDs.
- Quantified "deployment maturity" and found stagnation over one year (no clear throughput/latency trend) and inconsistent 5G advantage over LTE (e.g., 5G throughput lower in 1 city; 5G latency lower than LTE in only 3/8 cities).
- Conducted the first "metamorphosis" lifecycle study of 5G—from initial rollout to maturity—tracking coverage, throughput, and latency while comparing against mature LTE and accounting for factors like device adoption and network load.
- Identified 4-phase 5G coverage evolution (slow → rapid → second slow → stable) and 4-phase downlink throughput evolution, showing that coverage and throughput phases do not necessarily align.
- Showed that 5G uplink throughput remains largely unchanged over the 4-year period, while 5G latency decreases starting mid-2022; also reconciled latency discrepancies across studies by contrasting TCP-based Speedtest RTT vs ICMP ping and validating via controlled measurements.
- Published 2 Conference paper and 1 Journal Paper.

***Multipath TCP (MPTCP) for Dual-band 802.11ad + 802.11 ac networks***, *NSF Funded*.

C
Python
TCP
MPTCP
Android
Kernel
tc
802.11

- Ported MPTCP source code to a commercial dual-band smartphone (802.11ad@60 GHz + 802.11ac@5 GHz) by integrating 15,000+ LOC into the Android Linux kernel; released the implementation publicly for reproducible experimentation.
- Built an experimental pipeline to measure **100 ms**-granularity throughput and on-device power draw, enabling a first look at both *performance* and *energy* of MPTCP on resource-constrained phones.
- Identified and implemented key system-level optimizations for Gbps MPTCP on smartphones (e.g., enabling GRO and CPU/SIRQ affinity to big cores), achieving near sum-rate performance across the two radios under ideal conditions.
- Quantified throughput–energy tradeoffs across packet-scheduling ratios and derived practical guidelines toward an energy-aware MPTCP scheduler for dual-band mmWave/legacy Wi-Fi phones.
- Released two custom kernels with MPTCP stack for **ASUS ROG II** and **Pixel 5**.

## Selected Publications

### In Conference Proceedings

2025 A First Large-Scale Study of Operational 5G Standalone Networks. In *Proc. of ACM CoNEXT*, 2025.

2025 5G Metamorphosis: A Longitudinal Study of 5G Performance from the Beginning. In *Proc. of ACM IMC*, 2025.

2024 On the Predictability of Fine-grained Cellular Network Throughput using Machine Learning Models. In *Proc. of IEEE MASS*, 2024.

2024 An Experimental Study of Low-Latency Video Streaming over 5G. In *Proc. of IEEE MeditCom*, 2024.

2023 Performance of Cellular Networks on the Wheels. In *Proc. of ACM IMC*, 2023.

2021 Multipath TCP in Smartphones Equipped with Millimeter Wave Radios. Proc. of ACM WiNTECH '21, 2021.

### Journal Articles

2025 X5G: An Open, Programmable, Multi-Vendor, End-to-End, Private 5G O-RAN Testbed With NVIDIA ARC and OpenAirInterface. *IEEE Transactions on Mobile Computing*, 2025.

2025 How mature is 5G deployment? A cross-sectional, year-long study of 5G uplink performance. *Computer Communications*, 2025.

## Professional Services/Awards

2026 COMCOM, Computer Networks, IEEE Transactions on Green Comm. and Networking, *Reviewer*.

2025 COMCOM, IEEE VTC, IEEE Transactions on Green Comm. and Networking, *Reviewer*.

2024 Computer Networks, IEEE IoT, IEEE ICC, IEEE WCNC, *Reviewer*.

2024 IEEE LANMAN, IEEE WOWMOM, *Web Chair*.

2023 IEEE WCNC, *TPC Member*.

2021 NSF Funded Student travel grant, *IEEE Mobicom*.

2019 NSF Funded Student travel grant , *POWDER Network and Wireless Week*.